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PIC4SeR

SENSORS, EMBEDDED SYSTEMS AND ALGORITHMS FOR SERVICE ROBOTICS (SESASR)

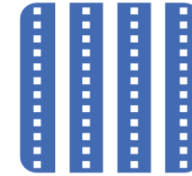
Lab03 – Hands-on Turtlebot3

Politecnico di Torino

M.Sc. Mechatronic Engineering



PRECISION
AGRICULTURE



SMART CITY
SEARCH &
RESCUE



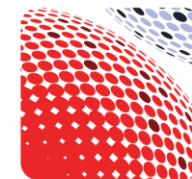
SERVICE
ROBOTICS
FOR
WELLBEING



CULTURAL
HERITAGE



UNDERWATER



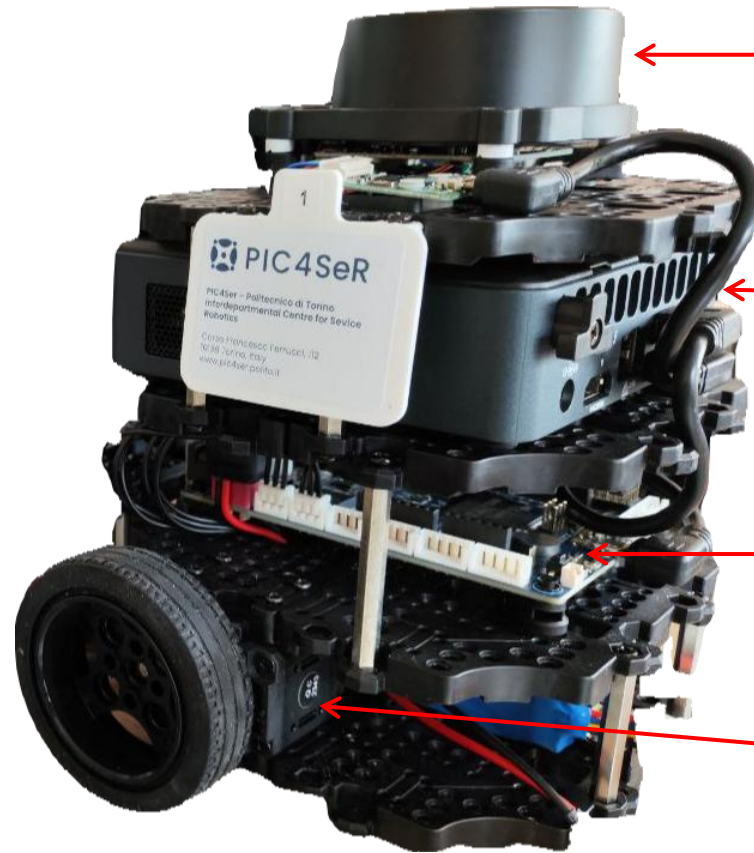
SPACE



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TURTLEBOT3 ANATOMY



2D LiDAR

Robot PC – NUC

Microcontroller board – OpenCR

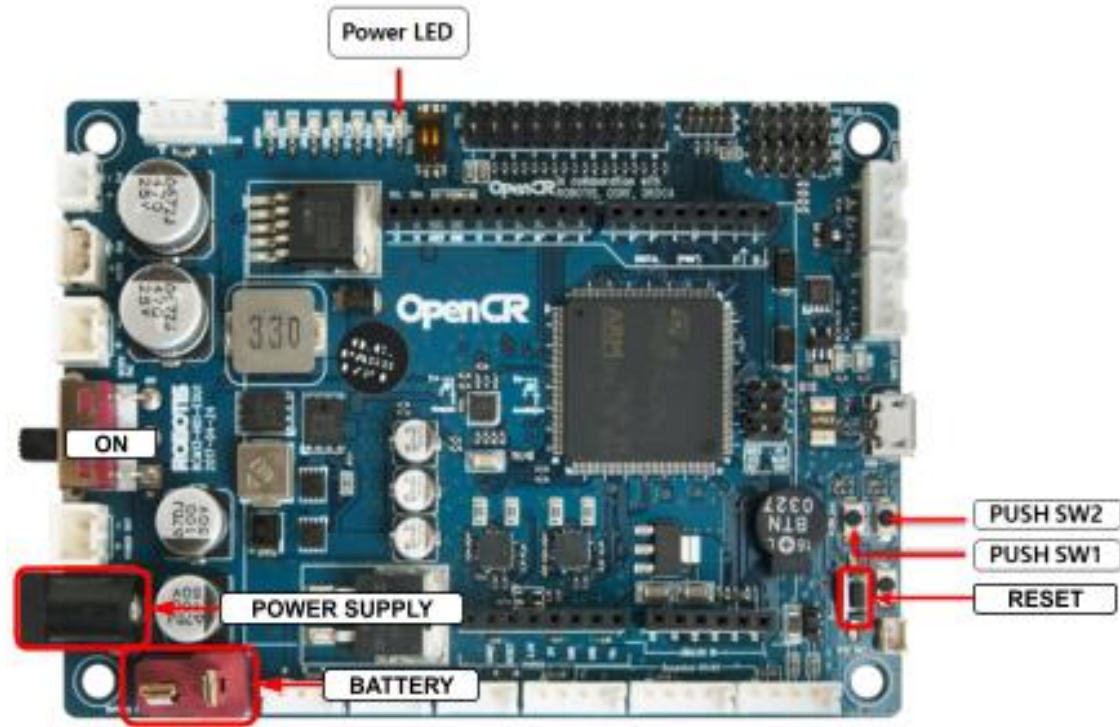
Motors – Dynamixel



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OPENCOR



- Motors control
- Sensors:
 - IMU
 - Wheel encoders
- USB communication



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ROBOT PC – NUC



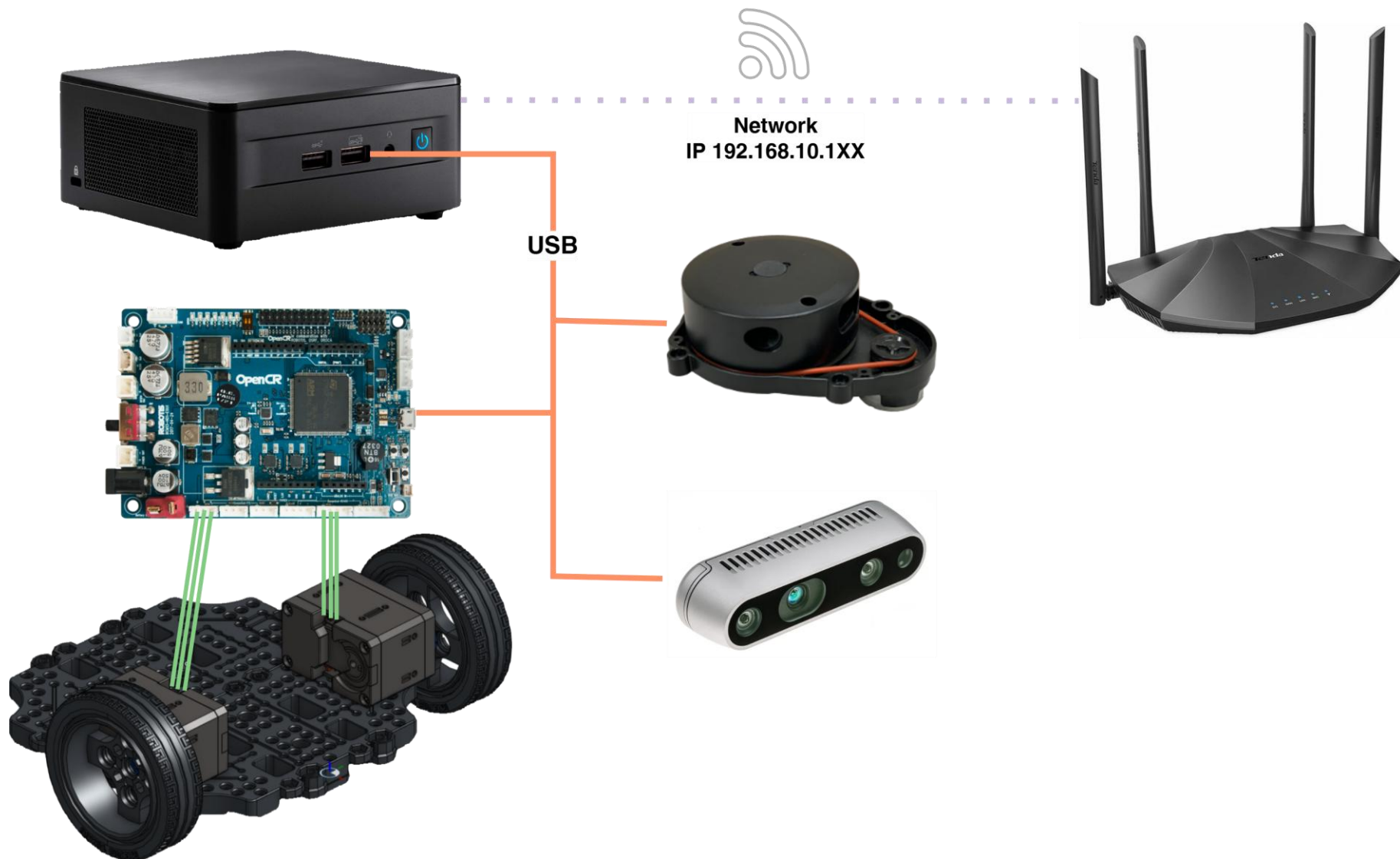
- Your interface to the robot
- Sensors drivers
- ROS 2 software



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ROBOT PC – NUC





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SSH – CONNECT TO ROBOT PC

- Open a terminal
- ssh [students@192.168.10.1XX](#)
- Now you are connected to a shell inside the robot PC

VS Code can be connected to the robot through SSH, check Lab instructions.



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ROS 2 – VISUALIZATION ON RVIZ

- Start the bringup on the robot
 - Connect through SSH
 - Export `RMW_IMPLEMENTATION=rmw_zenoh_cpp`
 - Export `ROS_DOMAIN_ID=X`
 - Export `TURTLEBOT3_MODEL=burger`
 - Export `LDS_MODEL=LDS-02`
 - `ros2 launch turtlebot3_bringup robot.launch.py`
- Start Rviz on your computer:
 - Export `RMW_IMPLEMENTATION=rmw_zenoh_cpp`
 - Export `ROS_DOMAIN_ID=X`
 - Export `ZENOH_CONFIG_OVERRIDE='mode="client";connect/endpoints=["tcp/192.168.10.1xx:7447"]'`
 - `ros2 launch turtlebot3_bringup rviz2.launch.py`